

Integrating Deep Learning into Clinical Decision-Making for Lumbar Degenerative Disease on MRI

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Inventory Category:	Adjacent lumbar MRI AI
Comparability / Priority:	Moderate Priority: Important

Study Details & Methodology from Literature Inventory

Study Type / MRI View:	Clinical comparison Lumbar MRI
Dataset & Cohort:	None (Sample: None)
Disease / Target:	Normal disc, disc herniation, spondylolisthesis
Model / AI Method:	PP-YOLOv2-based deep learning
Evaluation Metrics:	Sensitivity, specificity, accuracy
Key Finding / Relevance:	None
Thesis Context (Ch 2):	Clinical decision-support framing rather than model-only benchmarking.

Abstract / Summary

Introduction To develop an intelligent system based on artificial intelligence (AI) deep learning algorithms using deep learning tools, aiming to assist in the diagnosis of lumbar degenerative diseases by identifying lumbar spine magnetic resonance images (MRI) and improve the clinical efficiency of physicians. **Methods** The PP-YOLOv2 algorithm, a deep learning technique, was used to design a deep learning program capable of automatically identifying the spinal diseases (lumbar disc herniation or lumbar spondylolisthesis) based on the lumbar spine MR images. A retrospective analysis was conducted on lumbar spine MR images of patients who visited our hospital from January 2017 to January 2022. The collected images were divided into a training set and a testing set. The training set images were used to establish and validate the deep learning program's algorithm. The testing set images were annotated, and the experimental results were recorded by three spinal specialists. The training set images were also validated using the deep learning program, and the experimental results were recorded. Finally, a comparison of the accuracy of the deep learning algorithm and that of spinal surgeons was performed to determine the clinical usability of deep learning technology based on the PP-YOLOv2 algorithm. A total of 654 patients were included in the final study, with 604 cases in the training set and 50 cases in the testing set. **Results** The mean average precision (mAP) value of the deep learning algorithm reached 90.08% based on the PP-YOLOv2 algorithm. Through classification of the testing set, the deep learning algorithm showed higher sensitivity, specificity, and accuracy in diagnosing lumbar spine MR images compared to manual identification. Additionally, the testing time of the deep learning program was significantly shorter than that of manual identification, and the differences were statistically significant ($P < 0.05$). **Conclusions** Deep learning technology based on the PP-YOLOv2 algorithm can be used to identify normal intervertebral discs, lumbar disc herniation, and lumbar spondylolisthesis from lumbar MRI images. Its diagnostic performance is significantly higher than that of most spinal surgeons and can be practically applied in clinical settings.